

Asking a language model for three scores does not give you three measurements: a validity check for multi-axis text features, and what it costs to skip it

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Abstract

Practitioners increasingly prompt a language model for several conceptually distinct scores per document and build features that assume those scores measure different things. We show that this assumption can fail silently, give a cheap diagnostic that detects the failure before any model is fitted, and document what skipping it costs. The diagnostic is to name an external criterion for each axis in advance and check that each axis tracks its own criterion rather than its neighbour's; it requires no labelled data and, in our case, no additional model calls. We demonstrate on a three-axis decomposition of financial news into polarity, novelty and materiality, combined multiplicatively so that any near-zero factor vetoes the signal — a design we also characterise axiomatically, showing that continuity, strict monotonicity and ratio-scale separability force the combiner into the Cobb–Douglas family $\mathbf{s} \nu^\alpha \mu^\beta$ whose exponents the axioms leave free. Scoring 134,037 headlines for 35 US-listed companies, the diagnostic fails: novelty and materiality correlate at 0.871, and only materiality survives validation, predicting absolute market-adjusted returns at $t = 12.70$, while novelty carries no information about mechanical staleness beyond what materiality already carries. Two of the three axes are one latent judgement reported twice. The downstream consequences follow exactly: freely estimated exponents place the materiality elasticity at zero, the gated aggregate predicts returns no better than plain, count-weighted or relevance-filtered polarity, and adding it to a walk-forward selective forecaster moves precision at fixed coverage by between -1.2 and $+0.4$ percentage points

with every interval containing zero. The surviving axis measures magnitude rather than direction, which a signed product cannot use — a diagnosis available in advance from the validity check alone.

Keywords: Construct validity, Large language models, Feature engineering, Sentiment analysis, Financial text

1 Introduction

Instruction-tuned language models have made a particular move cheap: instead of extracting one score from a document, ask for several. A single prompt can request sentiment, urgency and credibility from a support ticket; severity, novelty and confidence from a clinical note; or, as here, polarity, novelty and materiality from a news headline. The scores come back well-formed, on the requested scales, and looking exactly like the distinct quantities that were asked for. Downstream they are treated as distinct: combined, weighted, interacted, used as separate features.

This paper is about a failure mode of that move. *Asking for k conceptually distinct scores does not guarantee k distinct measurements.* A model may return one latent judgement dressed in k labels, and nothing in the output reveals it: the scores vary, they correlate imperfectly, they respond sensibly to examples, and they are perfectly reproducible. The failure is silent, it is invisible in any internal consistency check, and it propagates into every feature built on top of it. Because the axes are correlated but not identical, even a careful analyst looking at a correlation matrix may conclude they are distinct-but-related rather than one thing measured twice.

The remedy we argue for is old and borrowed: *convergent validity*. Before building features, name for each axis an external criterion it must track if it measures what its label claims, then check that each axis tracks its own criterion rather than merely its neighbour's. The check needs no labelled data, and in our case it needed no additional model calls, since both criteria were computable from data already in hand. It is cheap enough to run first, and running it first would have saved the entire modelling exercise that follows.

We demonstrate on a concrete design from financial text analysis, chosen because it is one we built, believed in, and can therefore report against without charity. A large class of decision systems reads news and acts on it, scoring each item for sentiment polarity and averaging (Tetlock, 2007; Tetlock et al, 2008; Antweiler and Frank, 2004; Loughran and McDonald, 2011, 2016). Polarity answers only one question — is this good or bad? — and is silent on two others any desk asks reflexively: is the item *new*, or a restatement of something priced weeks ago; and does it *bear on value*, or is it cosmetic? Both omissions are documented to matter: investors underreact to stale news that recycles old information (Tetlock, 2011), and separating relevant from irrelevant news sharply changes the measured information content of a news flow (Boudoukh et al, 2019).

This paper takes a textual event to have three separable properties, each acting on price through a different channel: *polarity*, the sign and size of the implied value

change; *novelty*, how much genuinely new information the item carries relative to the market’s prior; and *materiality*, how sensitive fundamental value is to the event, independent of sign or freshness. *Multiplicative information gating* combines them as a product, which behaves like a soft logical AND: a stale or immaterial item produces a signal near zero however extreme its polarity, so a single weak factor vetoes the event.

Who supplies the three scores? Dictionaries handle polarity passably (Loughran and McDonald, 2011) but novelty and materiality need world knowledge, which is why they have historically come from commercial data vendors. An instruction-tuned language model, prompted with fixed numerical anchors and memoised for determinism, supplies all three at scale without a licence — and it is exactly this convenience that makes the validity question urgent, because the cost of adding an axis has fallen to almost nothing while the cost of a spurious axis has not.

The validity check fails on this design, and the rest of the paper is the audit trail of what that failure costs. We race the product against plain polarity, filtered polarity, count-weighted polarity and an additive combiner; we estimate the exponents of the three axes freely rather than assuming them; and we fit a walk-forward selective forecaster with and without the gated signal. Every one of those results is consistent with, and predictable from, the diagnostic that could have been run at the outset.

The framework is coupled to a decision stage, because a forecaster operating near the efficient-market limit (Fama, 1970) is wrong about direction most of the time and therefore owes its user two things: a probability that means what it says, and the discipline to abstain when conviction is low (Chow, 1970; Geifman and El-Yaniv, 2017). The same veto logic that suppresses an immaterial headline at the text stage suppresses a low-conviction name at the decision stage.

Our contributions are as follows.

1. **A validity protocol for multi-axis language-model features** (Section 4.3). For each axis, name an external criterion in advance, then test whether the axis tracks its own criterion *conditional on the other axes*. The conditioning does the work: raw correlations with a criterion look reassuring even when one latent judgement drives every axis. The protocol needs no labelled data and, here, no additional model calls.
2. **Evidence that the failure is real, and a worked diagnosis** (Section 5.3). On 134,037 headlines for 35 US-listed companies, two of three axes collapse: novelty and materiality correlate at 0.871, and only materiality survives validation — it predicts absolute market-adjusted returns at $t = 12.70$, while novelty carries no information about mechanical staleness beyond what materiality already carries.
3. **A measurement of what skipping the check costs** (Sections 5.4–5.7). Freely estimated exponents put the materiality elasticity at zero; the gated aggregate predicts returns no better than plain, count-weighted or relevance-filtered polarity; and adding it to a walk-forward selective forecaster moves precision at fixed coverage by between -1.2 and $+0.4$ percentage points, every interval containing zero. Each result is predictable from the diagnostic alone.
4. **A characterisation of the design under test** (Section 3). We prove a veto bound (Proposition 1) and show (Proposition 2) that continuity, strict monotonicity, sign fidelity and ratio-scale separability force the combiner into the

Cobb–Douglas family $\text{sign}(s)|s|^{\alpha_0}\nu^\alpha\mu^\beta$. The axioms do *not* pin the exponents, which is what makes them estimable and the design falsifiable rather than definitional.

5. **Deployment evidence on calibration and abstention** (Section 5.8) from a live, append-only ledger of 621 resolved interval forecasts, including a diagnosed and repaired calibration failure. It concerns the decision stage and is kept strictly separate from the claims about text, which it cannot support.

A note on how to read this paper. We proposed the decomposition, built the instrument, believed in it, and then found against it. The framework is reported in full alongside the evidence that it does not pay, because a design this intuitive will be reinvented, and because the useful artefact is not the verdict but the check that reaches it in an afternoon rather than a year.

2 Related work

2.1 Text, tone and returns

Tetlock (Tetlock, 2007) linked pessimistic media tone to downward price pressure and subsequent reversal; Tetlock et al (2008) showed that negative language in firm-specific news predicts earnings and returns. Antweiler and Frank (2004) studied message boards, and Loughran and McDonald (2011) demonstrated that finance-specific word lists matter because general-purpose ones systematically mislabel domain terms, a point developed in their later survey (Loughran and McDonald, 2016). Garcia (2013) showed the effect of tone is state-dependent, concentrated in recessions. Supervised and neural text models improve on dictionaries (Ke et al, 2019; Huang et al, 2023), and machine learning is now standard in empirical asset pricing (Gu et al, 2020). What unites this literature is that the per-item output is a scalar polarity; novelty and materiality, when handled at all, are handled by ad hoc pre-filters.

2.2 Relevance and novelty are established constructs, not new ones

We want to be explicit about prior art, because the decomposition proposed here is not the first attempt to look past polarity. Commercial news analytics have scored relevance and event novelty for well over a decade: RavenPack’s feeds attach a relevance score, an event novelty score and an event sentiment score to each item, and a large empirical literature filters on exactly those fields before running its tests. On the academic side, the novelty axis has a canonical antecedent in Tetlock (2011), who separates genuinely new stories from reprints of stale information and finds that investors react to the stale component; Boudoukh et al (2019) show that identifying which news is *relevant* materially changes measured information content; Dang et al (2015) study commonality in news flow across markets; and Kelly et al (2021) treat which text gets written at all as an object of study. Hillert et al (2014) document how media coverage intensity interacts with momentum, and DellaVigna and Pollet (2009) show that the same information moves prices differently depending on when attention is available to receive it.

Against that background we make no claim to have discovered novelty or materiality as constructs. Our residual contributions are narrower and, we think, defensible: (i) the three axes are produced *zero-shot by a general-purpose language model under a published prompt*, which replaces a licensed proprietary feed with something any research group can reproduce and audit; (ii) they are combined by a *continuous product with estimated elasticities* rather than used as hard pre-filter thresholds, which is how practitioners overwhelmingly use relevance scores; and (iii) the functional form is derived rather than asserted, and then tested against the alternatives on the same data.

2.3 Language models as measurement instruments

Instruction-tuned models emit structured numeric judgements from rubric-style prompts (Brown et al, 2020), and finance has adopted them quickly for return prediction (Lopez-Lira and Tang, 2023) and for reading corporate policy (Jha et al, 2024). Their self-reported confidence, however, is imperfectly calibrated and is perceived by users as better than it is (Steyvers et al, 2025). This motivates a deliberate division of labour in our design: we never ask the model to forecast. We ask it only to *decompose* an item along three defined axes, a narrower and more checkable task, and we leave forecasting and uncertainty quantification to a separate stage that can be calibrated against outcomes.

2.4 Calibration and selective prediction

Modern classifiers are frequently miscalibrated (Guo et al, 2017). Standard remedies are Platt scaling (Platt, 1999) and isotonic regression (Zadrozny and Elkan, 2002), assessed by expected calibration error and the Brier score (Brier, 1950). Stacking (Wolpert, 1992) complicates this by creating more than one score distribution, a failure mode we diagnose and repair in Section 4.6. Abstention is an old idea (Chow, 1970) with a modern theory (Geifman and El-Yaniv, 2017; El-Yaniv and Wiener, 2010) and a distribution-free relative in conformal prediction (Angelopoulos and Bates, 2023). Financial machine learning remains dominated by accuracy and area-under-the-curve reporting, which is fragile under backtest overfitting (Bailey et al, 2017) and multiple testing (Harvey et al, 2016; Romano and Wolf, 2005); comparatively little of it reports whether the probabilities can be trusted.

3 The multiplicative information model

3.1 Events, axes and the event signal

Let e denote a textual event – in our data, a news headline – about an entity, a listed equity. The event carries three scores:

- polarity $s(e) \in [-1, 1]$, the sign and size of the implied value change;
- novelty $\nu(e) \in [0, 1]$, the surprise relative to the market’s prior;
- materiality $\mu(e) \in [0, 1]$, the sensitivity of fundamental value to the event, independent of sign and of freshness.

The *event signal* is their product,

$$a(e) = s(e)\nu(e)\mu(e) \in [-1, 1]. \quad (1)$$

Define *relevance* $r(e) = \nu(e)\mu(e)$, so that $a(e) = s(e)r(e)$. The economic reading is a first-order decomposition: the expected price move is the product of a direction, a sensitivity and a surprise magnitude. Additive sentiment retains only the first, which is why it overreacts to loud but stale or immaterial items.

3.2 Aggregation

To aggregate the events $\mathcal{E}$ touching one entity in one period we use a relevance-weighted mean, restricted to events clearing a materiality floor μ_0 :

$$A = \frac{\sum_{e \in \mathcal{E}: \mu(e) \geq \mu_0} r(e) a(e)}{\sum_{e \in \mathcal{E}: \mu(e) \geq \mu_0} r(e)} = \frac{\sum r(e)^2 s(e)}{\sum r(e)}. \quad (2)$$

Relevance therefore enters twice, once inside the event signal and once as the weight, so a novel and material item dominates quadratically while cosmetic chatter is suppressed. We set $\mu_0 = 0.15$, the value used by the deployed system; Section 5.9 reports sensitivity to it.

3.3 Two properties of the product form

Proposition 1 (Veto bound). *For every event e , $|a(e)| \leq \min\{|s(e)|, \nu(e), \mu(e)\}$. In particular $a(e) = 0$ whenever any single factor is zero, and the aggregate of Eq. (2) satisfies $|A| \leq \max_e |s(e)|$.*

Proof. Each factor has modulus at most one and $\nu, \mu \geq 0$, so $|a(e)| = |s(e)|\nu(e)\mu(e)$ is a product of three numbers of modulus at most one; such a product cannot exceed the modulus of any one of them, which gives the per-event bound and the zero case. For the aggregate,

$$|A| \leq \frac{\sum_e r(e)^2 |s(e)|}{\sum_e r(e)} \leq \max_e |s(e)| \cdot \frac{\sum_e r(e)^2}{\sum_e r(e)} \leq \max_e |s(e)|,$$

where the last step uses $\sum_e r(e)^2 \leq (\max_e r(e)) \sum_e r(e) \leq \sum_e r(e)$ because every $r(e) \in [0, 1]$. The weights are non-negative but sub-stochastic, so the aggregate contracts the polarities toward zero and never amplifies them. $\square$

Proposition 1 is what “a single weak factor vetoes the event” means formally: a stale ($\nu \rightarrow 0$) or immaterial ($\mu \rightarrow 0$) item is capped near zero however extreme its polarity. No additive combination has this property, since a weighted sum $\alpha s + \beta \nu + \gamma \mu$ is generally nonzero when one term vanishes.

The second question is why a product rather than any other veto-respecting form. An earlier version of this argument assumed that scaling any one axis by λ scales

the combiner by λ , which is close to assuming the conclusion. The following statement replaces that assumption with the standard conditions of conjoint measurement (Debreu, 1960; Gorman, 1968; Krantz et al, 1971) plus the observation that each axis is measured on a ratio scale.

Proposition 2 (Representation). *Let $f : [-1, 1] \times (0, 1] \times (0, 1] \rightarrow [-1, 1]$ satisfy*

- (A0) normalisation: $f(1, 1, 1) = 1$;
- (A1) sign fidelity: $\text{sign } f(s, \nu, \mu) = \text{sign}(s)$;
- (A2) veto: $f = 0$ whenever any of $|s|, \nu, \mu$ equals 0;
- (A3) regularity: f is continuous and, on the interior, $|f|$ is strictly increasing in each of $|s|, \nu, \mu$;
- (A4') ratio-scale separability: for each argument there is a function h of one variable such that multiplying that argument by $\lambda \in (0, 1]$ multiplies $|f|$ by $h(\lambda)$, whatever the values of the other two arguments.

Then there exist exponents $\alpha_0, \alpha, \beta > 0$ with

$$f(s, \nu, \mu) = \text{sign}(s) |s|^{\alpha_0} \nu^\alpha \mu^\beta. \quad (3)$$

Proof. Work on the positive orthant $|s|, \nu, \mu \in (0, 1]$ and write $F = |f|$. Fix the second and third arguments. By (A4') there is h_0 with $F(\lambda x, \nu, \mu) = h_0(\lambda)F(x, \nu, \mu)$ for all admissible λ, x . Applying this twice,

$$h_0(\lambda_1 \lambda_2)F(x, \nu, \mu) = F(\lambda_1 \lambda_2 x, \nu, \mu) = h_0(\lambda_1)F(\lambda_2 x, \nu, \mu) = h_0(\lambda_1)h_0(\lambda_2)F(x, \nu, \mu),$$

and since $F > 0$ on the interior, h_0 satisfies the multiplicative Cauchy equation $h_0(\lambda_1 \lambda_2) = h_0(\lambda_1)h_0(\lambda_2)$. By (A3) h_0 is continuous and positive, so its only solutions are the power functions $h_0(\lambda) = \lambda^{\alpha_0}$ (Aczél, 1966), with $\alpha_0 > 0$ because F is strictly increasing in its first argument. Setting $x = 1$ gives $F(\lambda, \nu, \mu) = \lambda^{\alpha_0}F(1, \nu, \mu)$. Repeating the argument for the second and third arguments yields exponents $\alpha, \beta > 0$ and $F(|s|, \nu, \mu) = |s|^{\alpha_0} \nu^\alpha \mu^\beta F(1, 1, 1)$, and (A0) fixes $F(1, 1, 1) = 1$. Restoring the sign by (A1) gives Eq. (3). Continuity extends the representation to the closed domain, where (A2) is then automatic because all three exponents are strictly positive. $\square$

Remark 1. Proposition 2 delivers the *family*, not a member of it. The combiner of Eq. (1) is the special case $\alpha_0 = \alpha = \beta = 1$, and nothing in (A0)–(A4') selects it. This is deliberate. The axioms say that the three axes act on separate multiplicative scales and cannot compensate for one another, which is the substantive claim; the elasticities with which they act are an empirical matter, and Section 5.6 estimates α and β freely and tests $H_0: \alpha = \beta = 1$ rather than assuming it. Proposition 1 holds for every member of the family in the weaker form $|f| \leq \min\{|s|^{\alpha_0}, \nu^\alpha, \mu^\beta\}$, so the veto survives whatever the exponents turn out to be.

4 Data and methods

4.1 Two separate bodies of evidence

The paper draws on two datasets that are kept strictly apart, because they answer different questions and come from different markets.

A live deployment ledger (Indian large caps).

The forecasting system described in Sections 4.6 and 4.7 is deployed on liquid National Stock Exchange of India (NSE) large-capitalisation equities. Since April 2026 it has written every issued forecast to an append-only ledger – entity, timestamp, target date, point forecast, interval, nominal confidence, directional probability and horizon – with each row later stamped with the realised price and outcome. Rows are scored at first look and never retro-edited. This ledger supplies the evidence on interval coverage and calibration in Section 5.8. It cannot speak to the value of the text decomposition, because it stores no per-event return labels and its history is far too short to fit anything.

A large historical news panel (US equities).

To test the decomposition itself we need a long panel with dense per-item news and clean price data. We use FNSPID (Dong et al, 2024), a public corpus of time-stamped financial news headlines keyed to tickers. From the subset of symbols with usable price histories we take the SYMBOL_UNIVERSE best-covered names and remove exchange-traded funds, for which “the sensitivity of this firm’s fundamental value to the event” has no clean meaning. That leaves N_COMPANIES operating companies, deliberately close in number to the 54-name universe of the NSE deployment. Prices are split- and dividend-adjusted daily bars; returns are market-adjusted using per-symbol betas against SPY fitted on the training window only, and a Parkinson range estimator (Parkinson, 1980) supplies a volatility proxy.

We state the obvious limitation at the outset rather than in a closing caveat: *the validation market is not the deployment market*. There is no historical Indian headline corpus of comparable density available to us, so the question “does multiplicative gating carry information?” is answered on US data, and the question “does a calibrated selective forecaster behave as designed in production?” is answered on Indian data. Neither result is used to support a claim about the other market.

4.2 Scoring the three axes

The axes are produced by an instruction-tuned large language model prompted as a quantitative analyst and constrained to return a JSON array of $\{\nu, \mu, s\}$ triples, one per headline, in input order. Three design choices make the measurement reproducible.

First, the prompt fixes *numerical anchors* on every axis. For novelty, 0.0 is an exact restatement of news priced weeks ago or a generic market list, 0.2 an in-line and widely expected result, 0.6 an unexpected but modest development and 1.0 a sudden shock such as a surprise regulatory action. For materiality, 0.0 is boilerplate or a price-move round-up, 0.4 a contract or ruling worth a few percent of revenue, 0.8 a transformative

deal or guidance reset and 1.0 an existential event. For polarity the anchors run from -1.0 (fraud allegation) through 0.0 (a neutral reshuffle) to $+1.0$ (a large earnings beat or a takeover at a premium). Anchoring matters more here than in ordinary sentiment work because the product in Eq. (1) amplifies disagreement on any single axis.

Second, every scored item is memoised in a content-addressed store keyed by the SHA-1 hash of the entity and the normalised headline, so repeated runs are deterministic and the full scoring history is auditable.

Third, the axes are scored *zero-shot*: no labelled training data, no fine-tuning and no market outcome enters the prompt, so the scores cannot encode look-ahead information about the returns they are later used to predict.

The deployed Indian system uses Claude Haiku 4.5 for this role. The US validation panel was scored with Gemini 3.1 Flash-Lite. Using different models in the two settings is not ideal for comparability, but it does supply evidence on a question that matters for the protocol’s portability, namely whether the anchored prompt is model-specific; Section 5.1 reports test–retest reliability of the axes.

4.3 A validity protocol for multi-axis scores

Before any feature is built from the three axes, we state the check that this paper argues should come first. It is an application of convergent and discriminant validity, standard in psychometrics and unusual in applied text-as-data work, adapted to the situation where the “instrument” is a prompt.

1. **Name a criterion per axis, in advance.** For each axis, state an observable quantity that the axis must track *if* it measures what its label claims, and that the other axes have no particular reason to track. The criterion must be computable without the axis itself. Here, novelty is assigned mechanical staleness — lexical overlap with the same entity’s recent coverage — and materiality is assigned outcome magnitude, the absolute market-adjusted return.
2. **Test conditionally, not marginally.** Regress each criterion on all axes jointly, or equivalently examine partial correlations. This is the step that carries the diagnostic weight. If a single latent judgement drives every axis, each axis will show a respectable *marginal* association with every criterion, and only the conditional test reveals that none of them adds anything to the others.
3. **Inspect the shape before summarising it.** A rank correlation is meaningless if the relationship is not monotone. We plot each axis against deciles of its criterion first, and only then report a coefficient.
4. **Control the mechanical confounds of the criterion.** A criterion computed from a comparison set inherits that set’s properties; a document with few predecessors scores as novel by construction. The size of the comparison set enters as a control.
5. **Check that the criterion is the right *kind* of quantity for the downstream task.** An axis can be valid and still useless: one that legitimately measures magnitude cannot sharpen a directional forecast. This step is what turns a validity result into a design decision.

Two properties make this cheap enough to be worth doing routinely. Neither criterion here required labelled data, and neither required additional model calls: mechanical staleness is computed from the corpus already collected, and outcome magnitude from prices already needed for the study. Note also that the outcome criterion is used to validate the instrument and never as an input to any signal — checking whether a measurement means what it says against an outcome is legitimate, whereas building the measurement from that outcome would not be.

4.4 Aggregators raced against the product

For every symbol-session with at least one headline we compute A from Eq. (2) together with four competing aggregators and two ablations, all built from the same scored events so that no comparison is confounded by differences in the underlying text:

Mean polarity $\bar{s}$, the standard additive recipe.

Relevance-filtered polarity the mean polarity of events clearing the materiality floor. This is the baseline that matters most, because it is what practitioners actually do with a vendor relevance score, and it captures much of the same information as the product.

Count-weighted polarity $\sum_e s(e)/\sqrt{|\mathcal{E}|}$, which rewards repeated coverage as a naive “sum the sentiment” pipeline implicitly does.

Additive combiner the mean of $(s + \nu + \mu)/3$: the natural non-multiplicative way to use exactly the same three inputs.

Single-axis ablations the same relevance-weighted aggregate built from $s\nu$ alone and from $s\mu$ alone, which isolate how much of the product’s behaviour comes from each gate.

4.5 Timing convention

Every headline carries a UTC publication stamp. It is converted to America/New_York and assigned to the session whose 16:00 ET close first follows it, so the aggregate for session d contains only information public before that close and is used to predict returns realised from session $d + 1$ onward. This convention is applied uniformly and is the single most important guard in the study: a related analysis in our own earlier work produced a spuriously significant result that turned out to be a time-zone join artefact, and the convention above was adopted after that diagnosis.

4.6 From gated signal to calibrated probability

The aggregate A is one feature among several entering a forecaster that also sees price, momentum, trend, reversal and volatility features, and that emits per name and horizon a point forecast, a price interval and a directional probability $\hat{p}$. We treat the forecaster as a black box and study its calibration, measured by the expected

calibration error over B equal-width bins,

$$\text{ECE} = \sum_{b=1}^B \frac{|G_b|}{N} |\bar{y}_b - \bar{p}_b|, \quad (4)$$

with $\bar{p}_b$ and $\bar{y}_b$ the mean predicted probability and mean realised outcome in bin b , and by the Brier score $N^{-1} \sum_i (\hat{p}_i - y_i)^2$ (Brier, 1950).

The first deployed version of this model reported next-day up-probabilities clustered near 0.80 against a realised accuracy near 0.51. The cause was a stacking domain mismatch (Wolpert, 1992): an isotonic calibrator had been fitted on the out-of-fold *average of the base learners* and was then applied to the *stacked* prediction, a different distribution, and the blend itself was never calibrated. The repair is a three-stage procedure – refit the isotonic map on the base-average it was meant to correct, form the blend, then fit a final Platt scaler (Platt, 1999) on a held-out fold and apply it to the test and live rows. The general lesson is that a calibrator is bound to the distribution it was fitted on, and stacking silently creates more than one.

4.7 The conviction gate

Calibration makes $\hat{p}$ honest, not skilful. If short-horizon direction is close to unpredictable, a calibrated probability sits near the base rate almost everywhere and acting on every name is pointless. The remedy mirrors the text-stage veto: act only when conviction clears a threshold. For a horizon of H sessions we emit an UP call when $\hat{p} \geq \tau^*$ and abstain otherwise, choosing τ^* on training data as the smallest τ whose fired bucket reaches a target precision while still firing at least a minimum fraction of the time,

$$\tau^* = \min\{\tau : \text{prec}(\tau) \geq \pi_0 \text{ and } \phi(\tau) \geq \phi_0\}, \quad (5)$$

where $\text{prec}(\tau) = \Pr[y = 1 \mid \hat{p} \geq \tau]$ and $\phi(\tau) = \Pr[\hat{p} \geq \tau]$, with $\pi_0 = 0.60$ and $\phi_0 = 0.05$. The gate is one-sided because confident DOWN calls are destroyed by the upward drift of equities, so the system emits long-or-neutral signals only.

Reporting a single operating point invites the objection that τ^* was tuned to the number being reported. We therefore also report the full *risk-coverage curve* and its area (AURC): sorting predictions by confidence and sweeping the threshold traces the selective risk accepted at each coverage level, and the area under that curve summarises the whole trade-off without reference to any threshold (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017).

4.8 Evaluation protocol and inference

All panel results are walk-forward by calendar year: for test year Y the model sees only sessions strictly before 1 January Y , the last 20% of the training sessions are held out to fit the probability calibrator and to choose τ^* , and neither is ever selected on test data. Inference respects the two dependence structures of a return panel: a market-wide shock hits every name on the same date, and returns are persistent within a name. We therefore report two-way clustered standard errors by date and symbol (Cameron

et al, 2011), cross-checked against Fama–MacBeth cross-sectional slopes with Newey–West correction (Fama and MacBeth, 1973; Newey and West, 1987). Multi-session horizons are evaluated on non-overlapping windows as the primary specification, and confidence intervals for differences between models come from a block bootstrap that resamples whole dates, preserving cross-sectional correlation.

5 Results

We report the results in the order a sceptical reader would want them: first whether the instrument measures anything stable, then whether the three axes are the separate things the model of Section 3 assumes, then whether the gated aggregate predicts returns, and only last whether it predicts them better than the simpler recipes it is meant to replace. The answers are, in order: yes, *no*, yes, and no.

Table 1 The two bodies of evidence. The validation panel answers whether the decomposition carries information; the deployment ledger answers whether a calibrated selective forecaster behaves as designed in production. They are drawn from different markets and are never used to support claims about each other.

	Validation panel	Deployment ledger
Market	US listed equities	NSE India large caps
Source	FNSPID headline corpus	Live append-only forecast ledger
Entities	35 operating companies	7 names (live), 54 (walk-forward)
Span	2010-01-05 to 2023-12-29	19 Apr 2026 to 12 Jun 2026
Sessions	3,521	–
Symbol-sessions	117,312	–
... with scored news	48,261	–
Headlines scored	134,037	–
Resolved forecasts	–	621
Question answered	Does gating carry information?	Is the deployed system honest?

5.1 The measurement instrument

Scoring 134,037 headlines for 35 companies over 2010-01-05 to 2023-12-29 yields axes with sensible central tendencies: mean novelty 0.189, mean materiality 0.157 and mean polarity 0.039, the last confirming the mild positive skew that the news-tone literature reports. Two properties of the instrument deserve comment before anything is built on it.

First, the model reports on a *coarse grid* rather than a continuum: the mass sits on multiples of 0.1, visible as the discrete bars of Figure 2(a). This is a property of anchored prompting, not an artefact of our aggregation, and it means the axes carry less resolution than their $[0, 1]$ range suggests.

Second, the scores are *deterministic by construction*. Every item is memoised under a hash of the entity and normalised headline, so the panel is exactly reproducible from the cache. That is a property of the memoisation, not of the model, so we also rescored a random subsample of 250 headlines with the cache bypassed and a *different*

instruction-tuned model, which tests the stronger and more useful claim that the anchored protocol is portable rather than model-specific. Agreement is moderate and, tellingly, uneven across the axes: the intraclass correlation is 0.779 for polarity, 0.727 for materiality and only 0.565 for novelty, with the event signal $a = s\nu\mu$ inheriting 0.601.

The ordering matters for how the rest of the paper should be read. Polarity, the axis with decades of instrument development behind it, is the one two independent models agree on most; novelty, the axis on which the framework’s distinctive claim rests, is the one they agree on least. Any true multiplicative effect is therefore being estimated through the noisiest available channel, and classical attenuation would bias the measured contribution of gating toward zero. We return to this in Section 6.4, because it is the most credible non-trivial explanation of the null results that follow. The subsample is small — 250 headlines, bounded by API quota rather than by design — so these coefficients are indicative rather than precise.

5.2 The veto works, and the axes are not separable

The gate does exactly what Proposition 1 promises. Table 2 shows that 56.3% of events fall below the materiality floor and are refused outright, and that 47.6% of symbol-sessions carrying news end up with an aggregate of exactly zero. The contraction is equally visible: the standard deviation of the gated aggregate is 0.054 against 0.154 for mean polarity, so gating removes about two-thirds of the cross-sectional dispersion of the signal. Figure 2(c) is the per-session version of the same statement, with almost every point lying below the 45-degree line. On its own terms, the mechanism is not in doubt.

The premise underneath it is. Multiplicative gating supposes that novelty and materiality are *separable* properties travelling through different channels. In the measured data they are very nearly the same variable: the correlation between ν and μ across all 134,037 events is **0.871**, and Figure 2(b) shows the joint mass collapsing onto the diagonal. An item the model judges novel is almost always an item it judges material, and conversely.

This single fact anticipates every result that follows. If two of the three axes carry nearly the same information, then the second gate can add almost nothing once the first has acted, the product cannot separate itself from simpler combinations of the same inputs, and any attempt to estimate distinct elasticities for ν and μ will be poorly identified. All three happen.

Two very different things could produce that correlation. It may be a fact about the *world* — genuinely new firm news usually *is* the consequential news — or a fact about the *instrument*, an inability of the model to hold two related concepts apart under one prompt. The two have opposite implications, so the next subsection settles the question rather than leaving it open.

5.3 Do the axes track anything outside themselves?

Each axis has an external criterion it must track if it measures what it claims, and neither criterion requires further model calls.

Novelty should track mechanical staleness. Following the stale-news literature (Tetlock, 2011), we measure how much a headline repeats what was already written about the same ticker: one minus the maximum TF-IDF cosine similarity to that ticker’s headlines over the preceding thirty days. The comparison set is strictly earlier in time, so the measure is causal and uses no returns. It is available for 99.9% of headlines, with a median of 38 prior documents.

Materiality should track outcome magnitude. Materiality is defined as the sensitivity of value to an event *regardless of direction*, so it should predict the absolute market-adjusted return of the session. We stress that this criterion is used only to validate the instrument and never as an input to any signal: checking whether a measurement means what it says against an outcome is legitimate, whereas building the measurement from that outcome would not be.

The results separate the two axes sharply, but not in the direction the framework needs.

Novelty fails its own criterion. Figure 1(a) plots both axes against deciles of mechanical novelty, and the two curves are the same curve: parallel, identically shaped, separated by a constant. The relationship is also non-monotonic, rising from 0.167 at the stalest decile to 0.230 at decile 4 and then falling to 0.093 at the most lexically novel decile — the model assigns its *lowest* novelty to the headlines that share least vocabulary with recent coverage. Controlling for materiality and for the size of the comparison set, novelty’s standardised coefficient on staleness is -0.033 ($t = -6.51$) against materiality’s -0.022 ($t = -3.91$): statistically distinguishable from zero, but essentially the same number, so novelty carries no staleness information that materiality does not carry equally. The partial correlations tell the same story — -0.014 for novelty given materiality, -0.014 for materiality given novelty — as does the dense subsample with at least twenty prior documents (-0.155 and -0.157 , identical to three decimals).

Materiality passes its own criterion, decisively. Over 130,828 events, a one standard deviation increase in materiality is followed by 54.3 additional basis points of absolute market-adjusted return ($t = 12.70$, clustered by date), while novelty enters the same regression with the *opposite* sign at -26.3 basis points ($t = -8.69$). Whatever common factor drives the 0.871 correlation, the materiality label is the one that aligns with a real external quantity; conditional on it, the novelty label points the wrong way (Figure 1(b)).

The verdict is therefore *instrument*, and specifically the novelty axis. The model is largely reporting a single latent “importance” judgement on both channels; the channel labelled materiality happens to align with event magnitude, and the channel labelled novelty does not measure novelty in the sense the framework requires.

One implication deserves emphasis because it reaches past this instrument. The valid axis, materiality, predicts the *magnitude* of the move and not its *direction* — which is exactly what its definition promises. But Eq. (1) multiplies it into a *directional* signal. A quantity that carries information about how large a move will be cannot improve a forecast of which way it will go, and that, rather than any failure of measurement, is the cleanest explanation for the materiality exponent estimating

at zero in Section 5.6. The decomposition would have a much better chance against a volatility or absolute-return target than against a signed one.

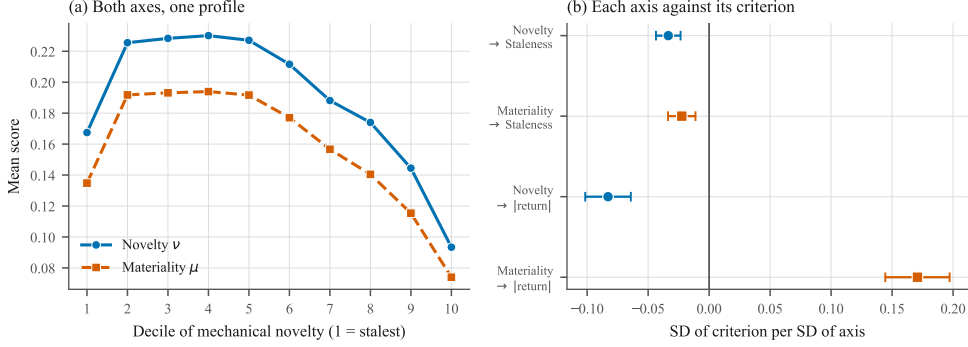


Fig. 1 Convergent validity of the two gates. (a) Mean novelty and mean materiality across deciles of mechanical novelty: the two curves are parallel and identically shaped, which is what one variable measured twice looks like, and both are non-monotonic in mechanical novelty. (b) Standardised coefficients of each axis on each criterion, in standard deviations of criterion per standard deviation of axis, with 95% intervals from date-clustered standard errors. Materiality predicts absolute return strongly; novelty does not track mechanical staleness beyond what materiality already tracks.

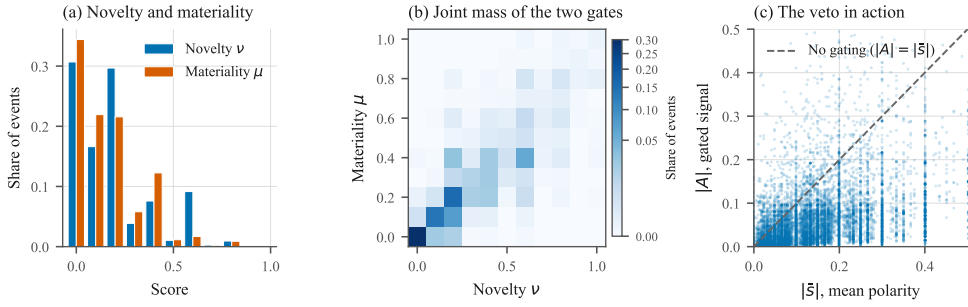


Fig. 2 The three axes and the veto they implement, measured on 134,037 scored headlines. (a) Marginal distributions of novelty and materiality; the model reports on a coarse grid, so the honest display is a discrete bar chart. (b) Joint mass of the two gates, on a power-law colour scale so the sparse high-novelty, high-materiality cells remain visible. (c) The gated session aggregate $|A|$ against the mean absolute polarity $|\bar{s}|$ of the same session: almost every point lies below the 45-degree line, which is Proposition 1 in the data.

5.4 Predictive content of the aggregators

The gated aggregate does predict returns. At the one-session horizon a one standard deviation move in A is followed by 12.2 basis points of market-adjusted return, with a two-way clustered t of 4.14 and a Fama–MacBeth t of 3.70; its rank information

Table 2 Descriptive statistics. Upper panel: the three axes and their products at the level of the individual headline. Lower panel: the session-level aggregators, computed on symbol-sessions carrying at least one scored headline. The final column is the share of exactly-zero values, which is the share of events or sessions the gate vetoes outright.

	Mean	SD	p25	Median	p75	Zeros (%)
<i>Panel A: per headline</i>						
Polarity s	0.039	0.207	0.000	0.000	0.200	52.0
Novelty ν	0.189	0.195	0.000	0.200	0.200	30.7
Materiality μ	0.157	0.166	0.000	0.100	0.200	34.4
Relevance $r = \nu\mu$	0.058	0.100	0.000	0.020	0.060	35.0
Event signal $a = s\nu\mu$	0.005	0.056	0.000	0.000	0.004	53.7
<i>Panel B: per symbol-session</i>						
Gated signal A	0.006	0.054	0.000	0.000	0.008	47.6
Relevance-filtered polarity	0.042	0.195	0.000	0.000	0.200	48.4
Mean polarity	0.038	0.154	0.000	0.000	0.100	36.8
Count-weighted polarity	0.058	0.227	0.000	0.000	0.200	36.8
Additive combiner	0.118	0.107	0.033	0.100	0.167	17.7
Novelty gate only ($s\nu$)	0.017	0.091	0.000	0.000	0.040	36.7
Materiality gate only ($s\mu$)	0.013	0.079	0.000	0.000	0.040	37.2

The correlation between novelty and materiality across all events is 0.871: the two gates are close to the same variable, which is the single fact behind most of what follows.

coefficient is 0.023, averaged over 2,548 trading days with a Newey–West t of 4.44. At five sessions the effect is 17.8 basis points ($t = 2.12$) on non-overlapping windows. By twenty-one sessions nothing survives: every aggregator has $|t| < 0.9$, which is what one expects of headline-level information at a monthly horizon.

So the text carries signal. The question this paper exists to answer, though, is whether the *product* carries more of it than the recipes it is meant to displace, and Table 3 says it does not. At one session, mean polarity earns a *higher* t -statistic than the gated aggregate (5.52 versus 4.14) on a nearly identical information coefficient (0.021 versus 0.023); count-weighted polarity produces a larger coefficient still (14.1 basis points) and the best information coefficient of the seven (0.023, $t = 4.72$). Relevance-filtered polarity — the hard-threshold recipe a practitioner would actually deploy — matches the product almost exactly (10.0 basis points, IC 0.023). The one aggregator that clearly underperforms is the additive combiner $(s + \nu + \mu)/3$ at 3.3 basis points and IC 0.010, which is worth stating plainly: *adding* the three axes is a bad idea, and to that extent the argument against additive combination in Section 3 survives. It is the step from a filter to a product that does not pay.

Figure 3 makes the overlap visible. At the horizons where anything is estimable, the confidence intervals of all seven aggregators sit on top of one another.

5.5 The nested horse race

Entering all four terms together (Table 4) was intended to be the sharp test: multiplicative gating predicts that the product loads and absorbs the lower-order terms. What happens instead is that nothing is individually identified. At one session the

Table 3 Predictive regressions of forward market-adjusted returns on each aggregator, one aggregator at a time. Regressors are standardised, so a coefficient reads as basis points of forward return per one standard deviation of signal. Standard errors are two-way clustered by date and symbol; horizons beyond one session use non-overlapping windows. Stars mark significance at the 10%, 5% and 1% levels.

Aggregator	$H = 1$		$H = 5$		$H = 21$	
	bps	t	bps	t	bps	t
Gated signal A	12.2***	4.14	17.8**	2.12	11.7	0.60
Relevance-filtered polarity	10.0***	5.11	12.5*	1.82	10.4	0.46
Mean polarity	9.3***	5.52	13.6**	2.03	18.5	0.87
Count-weighted polarity	14.1***	4.78	17.1**	2.29	14.3	0.62
Additive combiner	3.3***	3.82	3.9	0.71	-3.3	-0.16
Novelty gate only ($s\nu$)	11.1***	4.71	15.8**	2.14	7.3	0.38
Materiality gate only ($s\mu$)	11.9***	4.53	15.1*	1.89	10.8	0.60

Largest estimation sample: 48,260 symbol-sessions.

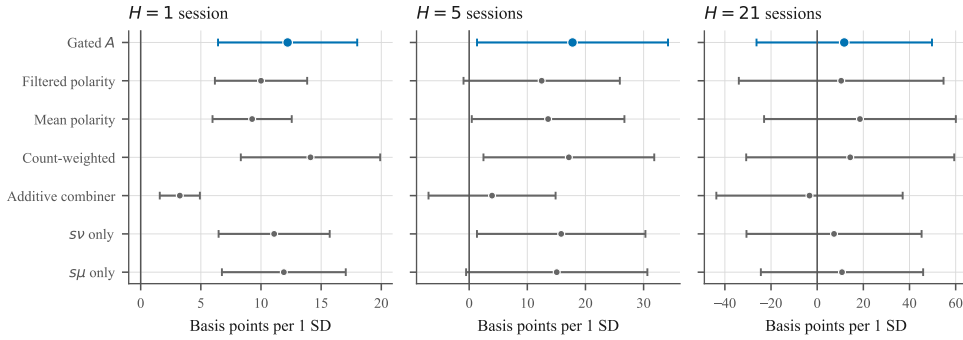


Fig. 3 Forward-return coefficient on each aggregator, in basis points per one standard deviation of signal, with 95% intervals from two-way clustered standard errors. The gated signal is highlighted; the alternatives are shown in neutral ink.

gated aggregate carries the largest coefficient (16.3 basis points) but only $t = 1.58$, while mean polarity falls to $t = 1.26$ and the two single-axis ablations take opposite signs. At five and twenty-one sessions the coefficients grow large and alternate in sign — +50.3 for the product against -36.2 for the materiality gate at five sessions — with $|t| < 2$ throughout.

This pattern is diagnostic of collinearity, not of a hidden effect, and Section 5.2 already told us why: with ν and μ correlated at 0.871 and all four regressors built from the same polarity, the design matrix is close to singular and the individual coefficients are not separately estimable. We therefore decline to read anything into the sign or size of any single term here. The honest conclusion from this table is a statement about what the data *cannot* answer: on this corpus, the nested horse race does not have the power to distinguish the product from its components.

Table 4 The nested horse race. All four terms enter one regression, so each coefficient is the marginal contribution of that term given the others. Multiplicative gating predicts that the gated signal loads and absorbs the lower-order terms. Regressors standardised; two-way clustered t -statistics.

Term	$H = 1$		$H = 5$		$H = 21$	
	bps	t	bps	t	bps	t
Mean polarity	6.2	1.26	23.1*	1.92	85.5*	1.76
Novelty gate only ($s\nu$)	-9.1	-1.30	-15.2	-0.63	-103.8*	-1.77
Materiality gate only ($s\mu$)	0.4	0.05	-36.2	-1.25	-64.7	-0.59
Gated signal A	16.3	1.58	50.3	1.33	111.8	0.98

5.6 Free exponents

The horse race cannot separate the terms, but the exponent grid can ask a cleaner question, because it varies the functional form directly instead of trying to identify collinear regressors. Remark 1 left α and β free; here we estimate them by sweeping $s\nu^\alpha\mu^\beta$ over $[0, 2]^2$ and reading off the information coefficient of the resulting aggregate.

The answer is unambiguous and it is not the one the unit-exponent product predicts. At one session the grid maximum sits at $\alpha = 0.75$, $\beta = 0.00$ with IC 0.0225: the materiality exponent is estimated at *exactly zero*. The unit-exponent product assumed by Eq. (1) scores 0.0210, and switching the gates off altogether — $\alpha = \beta = 0$, which collapses the aggregate to plain mean polarity — scores 0.0209. At five sessions the maximum is the no-gating corner itself, $\alpha = \beta = 0$ with IC 0.0156, against 0.0126 for the unit-exponent product: there, gating is not merely unhelpful but actively harmful.

Two conclusions follow. First, $H_0 : \alpha = \beta = 1$ is rejected, and rejected in the direction of the null model rather than toward some richer weighting. Second, and more important, the entire surface spans only 0.0201 to 0.0225 in information coefficient (Figure 4). The objective is nearly flat in the exponents, so the choice of gating weights — including the choice not to gate — moves predictive accuracy by an amount that is small relative to its own sampling error. That flatness, rather than the location of any maximum, is the finding, and it is why every comparison in Section 5.7 lands inside its confidence interval.

5.7 Selective forecasting with and without the gated signal

This is the experiment the framework has to pass, and the one earlier versions of this study did not run: the conviction gate is fitted twice on identical walk-forward folds, once on price features alone and once on price features plus the gated text signal, and the reported quantity is the difference between them.

There is no difference. Adding A moves precision at a fixed 10% coverage by +0.37 percentage points at one session (95% CI [-1.26, +1.96], $p = 0.685$), -1.23 points at five sessions ([-3.52, +1.13], $p = 0.322$) and +0.01 points at twenty-one ([-0.74, +0.72], $p = 1.000$). Against relevance-filtered polarity rather than against price alone the gaps are +0.07, -0.83 and +0.37 points. Every interval contains zero, and the point estimates change sign across horizons. The risk-coverage curves of Figure 5 say the same thing without reference to any threshold: the curves for the four feature sets are

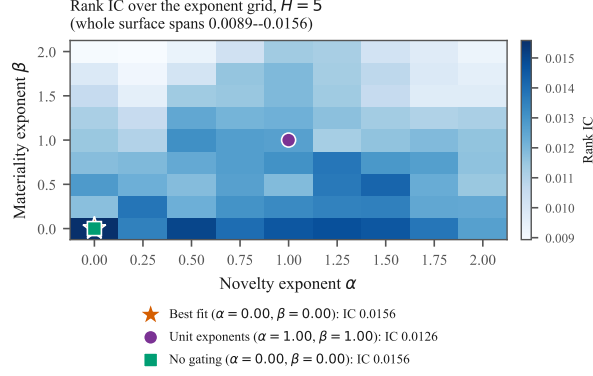


Fig. 4 Information coefficient of the aggregate built from $s\nu^\alpha\mu^\beta$ over a grid of exponents. The unit-exponent product assumed by Eq. (1) is marked, as is the empirical maximum; Remark 1 explains why the axioms leave this an open question.

indistinguishable over the whole coverage range, and their AURCs differ in the third decimal.

One result in Table 5 does not fit the pattern, and we report it because it is the most interesting thing in the table. The *text-only* model — no price features at all — achieves the best AURC at both short horizons (0.479 against 0.485 for price-only at one session, and 0.468 against 0.473 at five) and the best precision at five sessions (55.3% against 54.0% for price-only, on a 53.3% base rate). The headlines therefore carry decision-relevant information that the price features do not, and a learner given the text alone finds it. What the learner cannot do is exploit that information *once price features are also available*: the combined model is no better than price alone. Whether this reflects a genuine redundancy or a failure of the gradient-boosted learner to allocate capacity to a weak, sparse feature beside strong dense ones, we cannot determine from these data, and we would not want a reader to take the first explanation for granted.

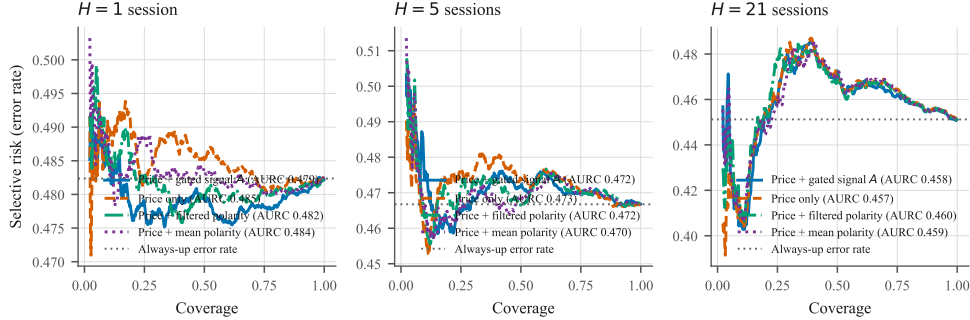


Fig. 5 Risk-coverage curves for the walk-forward selective forecaster. Each curve traces the error rate accepted at every coverage level, so the comparison does not depend on the choice of any single conviction threshold; the area under each curve (AURC, lower is better) appears in the legend.

Table 5 Selective forecasting with and without the gated text signal. Precision is the realised up-rate of the most confident 10% of out-of-sample predictions, a fixed coverage that makes the variants comparable without reference to any tuned threshold; AURC is the area under the risk–coverage curve, for which lower is better. All figures are pooled over walk-forward test years. The lower panel bootstraps the precision gap by resampling whole dates.

Feature set	$H = 1$		$H = 5$		$H = 21$	
	Prec.	AURC	Prec.	AURC	Prec.	AURC
Price + gated signal A	51.5	0.479	52.4	0.472	59.7	0.458
Price + filtered polarity	51.4	0.482	53.8	0.472	59.1	0.460
Price + mean polarity	51.2	0.484	53.4	0.470	59.3	0.459
Price only	51.2	0.485	54.0	0.473	59.0	0.457
Text only	52.8	0.479	55.3	0.468	58.9	0.462
Always-up base rate	51.8%		53.3%		54.9%	

Precision gap at 10% coverage	Estimate (pp)	95% CI	p
$H = 1$, vs. price only	+0.37	[-1.26, +1.96]	0.685
$H = 1$, vs. filtered polarity	+0.07	[-1.70, +1.83]	0.938
$H = 1$, vs. mean polarity	-0.05	[-2.23, +2.14]	0.948
$H = 5$, vs. price only	-1.23	[-3.52, +1.13]	0.322
$H = 5$, vs. filtered polarity	-0.83	[-3.10, +1.40]	0.482
$H = 5$, vs. mean polarity	-0.65	[-2.96, +1.64]	0.587
$H = 21$, vs. price only	+0.01	[-0.74, +0.72]	1.000
$H = 21$, vs. filtered polarity	+0.37	[-0.23, +0.97]	0.252
$H = 21$, vs. mean polarity	+0.28	[-0.32, +0.84]	0.345

5.8 Deployment evidence: coverage and calibration

The results above concern the text stage. The decision stage was studied on the deployed Indian system, where the evidence is of a different kind: not whether a signal predicts, but whether a system in production tells the truth about its own uncertainty.

It did not. Between 19 April and 12 June 2026 the system issued and later resolved 621 price-interval forecasts on seven NSE large caps, each carrying a nominal 90% interval with a median half-width of 5.6% of price. Had those intervals been calibrated, about 90% would have contained the realised price; empirical coverage was 70.7%, a 19.3 percentage-point shortfall (Table 6, Figure 6(a)). Most forecasts sat at the ten-day horizon, where coverage was 69.2% over 575 resolved forecasts. Nothing in the pipeline noticed, which is the point: overconfidence is silent unless somebody scores it.

The directional head failed in the same direction and for a diagnosable reason. It reported next-day up-probabilities clustered near 0.80 against a realised accuracy near 0.51, giving an expected calibration error of 0.345 and a Brier score above 0.366 — worse than predicting 0.5 every time. The cause was the stacking domain mismatch of Section 4.6, and the three-stage repair cut expected calibration error from 0.345 to 0.049 while leaving accuracy untouched. Calibration changed what the model

said about its confidence, not what it predicted, which is precisely the distinction the framework insists on.

With probabilities pulled back to the base rate, the walk-forward conviction gate on the 54-name deployment universe (2018–2026, 91,471 training rows and 50,272 out-of-sample rows) settles at $\tau^* = 0.63$ and fires on 5.6% of observations, abstaining on the rest. The fired bucket realises 60.6% directional precision against a 58.0% always-up base rate over 2,840 pooled calls: a 2.6 point edge with a 95% binomial interval of 58.8 to 62.4%. We do not over-read it. Per-year counts are small, the swing from +10.8 points in 2025 to -2.4 in 2024 is within sampling noise (Figure 6(b)), and the ranking area under the curve is about 0.47, so what edge exists lives in the high-conviction tail and not in the ordering. Crucially for this paper, *this gate contains no text feature at all*: it is evidence that calibrated abstention is worth building, not evidence for the decomposition.

Table 6 Deployment evidence from the live forecast ledger. Every interval carries a nominal 90% level. Panel B reports the walk-forward conviction gate on the 54-name deployment universe, where precision is the realised up-rate of the fired high-conviction bucket and the base is the unconditional always-up rate.

<i>Panel A: interval coverage by horizon</i>				
Horizon	Resolved forecasts	Nominal	Empirical	
5 trading days	8	90.0%	100.0%	
10 trading days	575	90.0%	69.2%	
20 trading days	38	90.0%	86.8%	
All	621	90.0%	70.7%	

<i>Panel B: walk-forward conviction gate by test year</i>				
Test year	Fires	Fired precision	Always-up base	Edge (pp)
2022	365	61.6%	54.0%	+7.6
2023	290	68.3%	65.3%	+3.0
2024	932	53.3%	55.7%	-2.4
2025	822	69.0%	58.2%	+10.8
2026	431	54.5%	47.1%	+7.4
Pooled	2,840	60.6%	58.0%	+2.6

The pooled base is the unconditional always-up rate over all out-of-sample rows, not the fires-weighted mean of the yearly bases (55.9%); the pooled edge is therefore the more conservative of the two comparisons.

5.9 Robustness

A null result is only worth reporting if it is not an artefact of the analyst’s choices, so we varied the three that matter.

The materiality floor. Sweeping μ_0 from 0 (no floor) to 0.40 leaves the one-session coefficient on A between 12.0 and 12.2 basis points with t between 3.97 and 4.14.

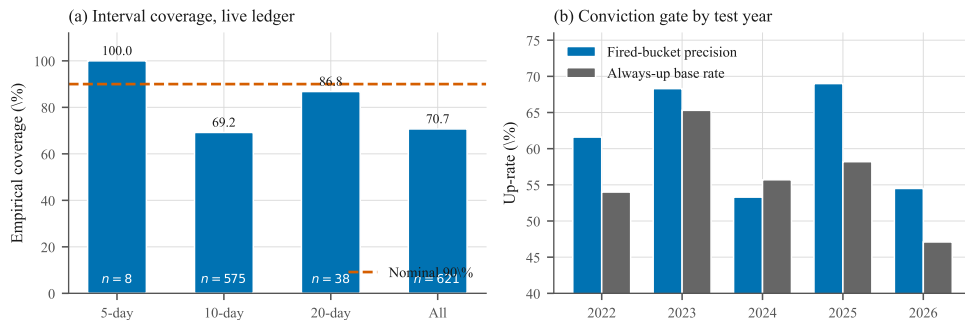


Fig. 6 Deployment evidence. (a) Empirical coverage of nominal 90% price intervals on the live ledger, by horizon. (b) Fired-bucket precision of the conviction gate against the always-up base rate, by walk-forward test year, at roughly a 5.6% firing rate.

The floor simply does not bind on the conclusion, which is what one expects when the exponent on materiality is estimated at zero. One detail cuts the other way and we note it: at five sessions the information coefficient rises from 0.013 at $\mu_0 = 0$ to 0.025 at $\mu_0 = 0.25$ and above, with the Newey–West t going from 1.24 to 2.25. A *hard relevance filter* thus helps where multiplicative weighting does not — consistent with the exponent surface, and consistent with what practitioners already do with vendor relevance scores.

Overlapping versus non-overlapping windows. Using every session rather than every H -th weakens the five-session result from 17.8 basis points ($t = 2.12$) to 8.9 ($t = 1.68$), in the direction that the overlapping-observation literature predicts. We report non-overlapping windows as the primary specification for that reason; neither choice changes the comparison between aggregators, which is what the paper turns on.

Quiet sessions. Restricting the panel to symbol-sessions that carry scored news is the primary specification, since including quiet sessions pads the sample with rows where every text aggregator is identically zero and can only dilute any difference between them.

Repeatability of the instrument. Cross-model agreement is reported in Section 5.1. We regard it as the weakest link in the measurement chain and the first thing a replication should attack.

6 Discussion

6.1 Why the decomposition did not pay

Section 5.3 lets us be specific rather than speculative, and the answer has two parts.

One axis is not measuring what it is named. Novelty fails its external criterion: it does not track mechanical staleness beyond what materiality already tracks, its profile against mechanical novelty is the same curve as materiality’s, and it is non-monotonic besides. Materiality, by contrast, passes decisively, predicting absolute market-adjusted returns at 12.70 standard errors. So the 0.871 correlation is not the world telling us that novel news is material news; it is one latent “importance”

judgement being reported twice under two labels, only one of which corresponds to something external. A product of two gates cannot behave like a two-gate product when only one gate exists.

The surviving axis carries the wrong quantity for the task. Materiality is by definition about magnitude irrespective of sign, and that is exactly how it behaves. Multiplying it into a *signed* signal, as Eq. (1) does, asks a magnitude variable to sharpen a direction forecast, which it has no means of doing. This is the cleanest reading of the zero materiality exponent in Section 5.6, and it points somewhere constructive: the same decomposition aimed at a volatility or absolute-return target, where materiality’s information is the relevant kind, is a genuinely different experiment and one we have not run.

Attenuation remains a live secondary concern. Cross-model intraclass correlations are 0.779 for polarity but only 0.565 for novelty, and the product inherits 0.601; multiplying noisy factors compounds their noise, so classical errors-in-variables would bias the measured contribution of gating toward zero. We cannot rule this out, and it is a further reason the framework deserves a rematch with a better instrument rather than a dismissal. What it does not license is treating the null as a measurement artefact by assumption — and in any case attenuation cannot explain why novelty points the *wrong way* on absolute returns, which is a validity failure and not a noise problem.

Beyond that, note what did survive. The additive combiner performed worst of all the aggregators, so collapsing the axes by addition is clearly wrong; and a *hard* materiality filter improved the five-session information coefficient from 0.013 to 0.025 where multiplicative weighting did not. Read together, these say that relevance information is worth using as a *filter* — decide which items to look at — but not as a continuous multiplicative *weight* on the ones that survive. The practitioner convention of thresholding a vendor relevance score, which we set out to improve on, appears to be the right call.

6.2 What this means for practice and for the literature

For a practitioner the recommendation is unglamorous and specific: filter on relevance, then average polarity. The elaboration we proposed costs an extra language-model call per headline, introduces a second noisy axis, and returns nothing measurable. Where a language model does earn its keep is upstream, in deciding which items to consider at all.

For the literature, the more useful contribution may be the measurement rather than the null. That a competent language model, given careful definitions and four numerical anchors per axis, returns novelty and materiality scores correlated at 0.871 — of which only the materiality channel survives an external validity check — is a fact about these instruments that anyone building multi-axis text features should know before designing around an assumed separation. Prompt-based scores are not independent merely because the prompt asks for independent things, and asking a model for k conceptually distinct scores does not guarantee k distinct measurements. The cheap diagnostic we would urge on anyone in this position is the one in Section 5.3: name an external criterion for each axis *before* building features from it, and check that each axis tracks its own criterion and not merely its neighbour’s.

We also note one asymmetry worth following up. The text-only model beat the price-only model on AURC at both short horizons, while the combined model beat neither. Weak, sparse, noisy features sitting beside strong dense ones are exactly the case where a gradient-boosted learner is known to underuse them, so the right conclusion is not that the text is redundant but that we have not shown how to combine it. That is a question about model architecture rather than about the decomposition, and it is where we would look next.

6.3 What the design buys, and what it costs

Every result in this paper turns on one trade-off. The text-stage veto refuses stale or immaterial items and the decision-stage gate stays out on most names, so the system gives up commenting on everything in exchange for the right to be believed on the little it flags. That trade is not free, and the risk-coverage curves of Figure 5 are the honest way to price it: they show what error rate is bought at every level of coverage rather than at the single operating point that happens to flatter the method.

Nothing in Section 3 is specific to equities. Any stream of timestamped textual events feeding a decision has the same three axes, and the same failure mode when they are collapsed into one score: triage of clinical alerts, operations tickets and content-moderation queues all conflate “how bad” with “how new” and “how consequential”. The decision-stage half of the design is an instance of selective classification (Chow, 1970; Geifman and El-Yaniv, 2017), here paired with a probability whose calibration was repaired rather than assumed, and the interval-coverage failure of Section 5.8 is exactly the problem distribution-free methods (Angelopoulos and Bates, 2023) are built for.

6.4 Limitations

Five limitations bound what may be concluded, and we would rather state them than have them found.

The validation market is not the deployment market. The decomposition is tested on US headlines and the deployed system trades Indian equities. We report no result that crosses that boundary, but the reader should not read the US panel as evidence about NSE microstructure, disclosure regimes or news supply, all of which differ.

The axes are judgements, and one of them did not survive validation. Anchoring the prompt tightens the scores and Section 5.1 quantifies their repeatability, but repeatability is not validity: a scorer can be perfectly consistent and consistently wrong, which is close to what Section 5.3 finds for novelty. Every result about gating is therefore a joint test of the decomposition and of this particular instrument, and a differently worded rubric, a larger model, or a two-stage prompt that scores the axes independently might separate them where ours did not. We consider that the single most valuable follow-up.

The external criteria are proxies. Lexical similarity over a thirty-day window is a serviceable but crude stand-in for what the market already knew, and absolute return is a noisy realisation of an event’s true importance. A criterion failure is therefore weaker evidence than a criterion success: materiality passing is strong evidence it

measures something, whereas novelty failing is consistent with both a bad axis and a bad proxy.

The comparison is against feasible baselines, not against a vendor feed. The most informative comparison would be against a commercial relevance and novelty product on the same headlines. We do not have a licence for one, so the relevance-filtered polarity baseline is our best available stand-in for what a practitioner would actually deploy.

The live ledger is young and small. Two months and seven names is a snapshot of coverage behaviour, not a stationary law, and the live directional sample is far too small to support a claim about skill. That is why the live evidence in this paper is confined to calibration and interval coverage, where 621 resolved forecasts is enough to say something, and the claims about selection rest on walk-forward evidence instead.

Selection and multiple testing. We report several horizons, several aggregators and several feature sets. The risk-coverage curves and the non-overlapping-window specification remove the most common sources of optimism, and differences between models carry bootstrap intervals, but a reader who wants a single number with a family-wise guarantee should apply a stepdown correction (Romano and Wolf, 2005) to the horse-race column and read the result accordingly. Financial machine learning has a well-documented tendency to discover edges that do not survive (Bailey et al, 2017; Harvey et al, 2016), and we would rather our estimates be read with that prior in mind.

7 Conclusion

A textual event carries direction, surprise and consequence, and there is a clean argument that these should combine multiplicatively: the product makes freshness and relevance preconditions for influence, it caps the signal at its weakest factor, and under continuity, monotonicity and ratio-scale separability it is the form the axes must take, up to exponents the axioms leave free.

Measured against real data, the argument does not pay. On 134,037 headlines covering 35 companies, the gated aggregate predicts market-adjusted returns at short horizons but no better than mean polarity, count-weighted polarity or relevance-filtered polarity; estimating the exponents freely puts materiality’s at zero and, at five sessions, collapses the aggregate to plain polarity; and adding the gated signal to a walk-forward selective forecaster moves precision at fixed coverage by between -1.2 and $+0.4$ percentage points, with every interval containing zero.

The explanation is in the measurement rather than in the market, and it is specific. Novelty and materiality, as an instruction-tuned model scores them, correlate at 0.871; validated against external criteria, only materiality survives, predicting absolute market-adjusted returns at 12.70 standard errors, while novelty fails to track mechanical staleness beyond what materiality already tracks and enters absolute returns with the wrong sign. The three-axis decomposition is empirically a two-axis one, and the axis that does measure something measures *magnitude*, which a signed product has no way to exploit.

Two things do survive. Adding the axes is worse than multiplying them, so the case against additive combination stands. And using relevance as a hard filter rather than a continuous weight improves the five-session information coefficient from 0.013 to 0.025 — which is what practitioners already do with vendor relevance scores, now with evidence behind it. Separately, on the deployment side, nominal 90% intervals covered 70.7% of live outcomes, a three-stage recalibration cut expected calibration error from 0.345 to 0.049 without touching accuracy, and a conviction gate abstaining on 94% of names raised the thirty-day hit rate to 60.6% against a 58.0% base — evidence that calibrated abstention is worth building, independent of anything the text stage does.

The framework is intuitive and will be proposed again. We think the more useful contribution is the record of what happened when it was built, instrumented and tested: the decomposition is measurable, the veto provably and visibly works, and it buys nothing over averaging polarity — for reasons that are now diagnosed rather than guessed at. Two experiments follow directly and neither is run here. An instrument whose novelty channel passes an external validity check would give the framework the fair test it has not yet had; and pointing the same decomposition at a volatility or absolute-return target, where materiality’s information is of the relevant kind, asks the question the signed formulation cannot. The measurements reported here are designed to make both possible, and to make it costly for the next version of this idea to be adopted without them.

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Declarations

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Competing interests

The authors declare no competing interests.

Availability of data and materials

The FNSPID news corpus is publicly available ([Dong et al, 2024](#)); price data are public end-of-day records. The scored axis values, the derived panel, the analysis code and the live forecast ledger underlying the reported tables and figures are available from the corresponding author on reasonable request.

Use of generative artificial intelligence

An instruction-tuned language model is used as a measurement instrument *within the system under study*, scoring headlines along the three axes from the anchored prompt of Section 4.2; those outputs are persisted for audit. During manuscript preparation a large language model was used solely to improve language and readability. No generative tool produced the data, results, figures or interpretations, and the authors take full responsibility for the content.

Authors' contributions

Both authors contributed to the study conception and design, analysis and interpretation. Both authors read and approved the final manuscript.

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